

Completing the square



- 1 Explain how to find the number to add to $x^2 - 5x$ to make it a perfect square trinomial.
- 2 Consider the expression $x^2 + 6x$.
 - a Does adding 36 to the expression make it a perfect square?
 - b What should be added to $x^2 + 6x$ to make it a perfect square?
- 3 Consider the equation $x^2 - 3x = 6$. What should be added to the equation in order to complete the square?
- 4 Complete the following expressions so they form a perfect square:
 - a $x^2 + 10x + \square$
 - b $x^2 - 2x + \square$
 - c $x^2 - 11x + 16$
 - d $x^2 - x + \square$
 - e $x^2 + mx + \square$
 - f $x^2 - \frac{4}{5}x + \square$
- 5 Complete the following perfect squares:
 - a $x^2 + 4x + \square = (x + \square)^2$
 - b $x^2 - 5x + \square = (x - \square)^2$
 - c $x^2 - \frac{7}{4}x + \square = (x - \square)^2$
- 6 Rewrite the following in the form $(x + b)^2 + c$ using the method of completing the square:
 - a $x^2 + 12x$
 - b $x^2 - 12x$
 - c $x^2 + 2x + 2$
 - d $x^2 - 14x + 56$
 - e $x^2 - 6x + 5$
 - f $x^2 + 3x + 6$
 - g $x^2 - 7x + 14$
- 7 Factorise the quadratics using completing the square:
 - a $y = x^2 + 56x + 159$
 - b $y = x^2 - 24x + 63$
- 8 Factorise the quadratics using the method of completing the square to get them into the form:
$$y = (x - a + \sqrt{b})(x - a - \sqrt{b})$$
 - a $y = x^2 - 6x + 7$
 - b $y = x^2 + 8x + 14$

Non-monic completing the square

- 9** Using the method of completing the square, rewrite $4x^2 - 36x + 79$ in the form $c((x + a)^2 + b)$.
- 10** Rewrite the following in the form $a(x + b)^2 + c$ using the method of completing the square:
- a** $3x^2 + 9x + 8$ **b** $4x^2 - 11x + 7$
- 11** Use completing the square to factorise the quadratic $y = 2x^2 + 25x + 23$.

Solving equations

- 12** Consider the equation $x^2 + 24x = 10$. To solve the equation by completing the square, state whether each of the following lines of working could be used:
- a $x^2 + 24x + 144 = 10$
- b $(x + 12)^2 - 144 = 10$
- c $(x + 12)^2 = 10$
- d $(x + 12)^2 - 144 = -134$
- e $x^2 + 24x + 144 = 10 + 144$
- 13** Solve the following quadratic equations:
- a $(x + 6)^2 = 4$
- b $(4x - 4)^2 = 4$
- c $2(x - 3)^2 = 8$
- d $(x + 2)^2 = 20$
- e $(x + 5)^2 - 2 = 15$
- f $(x + 9)^2 = \frac{15}{2}$
- 14** Solve the following quadratic equations by completing the square:
- a $x^2 + 18x + 32 = 0$
- b $x^2 - 6x + 8 = 0$
- c $x^2 - 9x + 8 = 0$
- d $2x^2 - 12x - 32 = 0$
- e $4x^2 + 11x + 7 = 0$
- f $x^2 - 2x - 32 = 0$
- 15** Solve the following quadratic equations by completing the square. Leave your answers in surd form:
- a $x^2 + 24x + 5 = 0$
- b $x^2 + 11x + 5 = 0$
- c $6x^2 + 48x + 24 = 0$
- d $x^2 - 7x + 8 = 0$
- e $x^2 + x - 2 = 0$
- f $5x^2 + 55x + 3 = 0$



Applications

16 A circle has equation $(x + 2)^2 + (y - 4)^2 = 41$. Find the x -intercepts of the circle.

17 Find the value(s) of k that will make $x^2 + kx + 16$ a perfect square trinomial.

18 The revenue y (in millions of dollars) of a company x years after it first started is modelled by

$$y = 12.5x^2 - 64x + 135$$

By completing the square, use this equation to predict the number of years x after which the revenue of the company will reach \$1167 million.